

AYUSHMAN PAUL

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EDUCATIONS

Indian Institute of Technology Kharagpur

Sep 2023 - Present

Dual Degree in Manufacturing Science and Engineering

CGPA : 8.17/10

Relevant Coursework : Data Structures & Algorithms, Machine Learning, Deep Learning, Probability & Statistics, Linear Algebra, Algorithms I, Transform Calculus.

WORK EXPERIENCES

AI/ML Intern | Part-time

Feb 2025 - Jun 2025

Nirveon X

Remote

- Developed a **GenAI-powered Mental Health Assistant** using **RAG pipelines** and **Agentic AI workflows**.
- Implemented **HyDE**, **GRAG**, and **LCM+RAG strategies** to enhance **response accuracy** and user empathy.
- Conducted research to integrate **advanced retrieval methods** within the agentic pipeline.

Technologies Used : GRAG, LangChain, HyDE

PROJECTS

NyayaAI | 🔄 Github

Dec 2025 - Jan 2026

Convolve 4.0 - Pan IIT Hackathon

- **Developed** a Multi-Agent legal assistance system to **democratize access** to Indian legal information.
- **Architected** specialized reasoning and knowledge agents with Web Search tools for **real-time, cited legal advice**.
- **Integrated** Qdrant for vector-based retrieval of legal precedents and built a **user-friendly Streamlit interface**.

Technologies Used : FastAPI, Qdrant, Groq

Multimodal Product Pricing Engine | 🔄 Github

Aug 2025 - Sep 2025

Amazon ML Challenge 2025

- Engineered a **high-precision multimodal price prediction system** fusing catalog text and image embeddings, optimizing for **SMAPE** on large-scale data.
- Implemented an **advanced ensemble regression pipeline** with robust feature extraction to handle **extreme price variance** and enhance generalization.

Technologies Used : NLP, Computer Vision, VLM

Project Kisan | 🔄 Github

Jun 2025 - Jul 2025

Google Cloud Agentic AI Hackathon

- **Architected** a multi-agent AI platform with a central **Gemini-powered router** to intelligently classify and delegate multimodal queries (voice, image, text).
- **Integrated** diverse APIs (**Google Maps**, **OpenWeatherMap**) for real-time crop diagnosis and market prices.
- **Ensured** accessibility with a voice-first, multilingual interface and an offline-first dashboard using **Firestore Local Cache**.

Technologies Used : LangGraph, LangChain, Gemini

Marketing Channels Budget Optimization | 🔄 Github

Feb 2025 - Mar 2025

General Championship Technology 2025

- **Secured Gold Medal** for Meghnad Saha Hall by developing an **algorithmic approach** to optimize marketing budget allocation.
- Analyzed conversion data across multiple channels to maximize **ROI** using **statistical modeling** and optimization techniques.
- Processed large datasets to visualize spending efficiency and predict future performance for **strategic decision-making**.

Technologies Used : Python, Jupyter Notebook

Research Paper Evaluation Framework | 🔄 Github

Dec 2024 - Jan 2025

Kharagpur Data Science Hackathon 2025

- Developed a **dual-task ML framework** for analyzing research papers and recommending suitable conferences.
- Integrated **citation density**, readability scores, and **SPECTER-based semantic similarity** for analysis.
- Built a **Random Forest classifier** (F1: 0.8) for publishability prediction and implemented LLM-based justification.

Technologies Used : Pathway, Streamlit, RAG

SKILLS

Programming Languages :

Python, C/C++, SQL, HTML, CSS, JavaScript

Frameworks & Libraries :

PyTorch, TensorFlow, LangChain, LangGraph, Fast API, Scikit-learn, Pandas, Django, MCP

Tools & Platforms :

Git, Docker, VS Code, Google Colab, Anaconda, Cursor, Google Antigravity, GitHub, GitLab